

Project Topic: Bookstore

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- **Short description:** In this project, I aim to make a functional virtual bookstore, where users can explore, select and purchase different book
- **Project scope:** The goal is to design a website that has the potential to be used as a medium to access a bookstore, online. Both users and employees will have access to the web, requiring log in. Users will be able to not only look at the collection but also purchase them, leave reviews and comments about the book. Similarly, bookstore employees will use the app to assist clients, update the library and its collection, check and change the availability status of the book collection and take the orders that are placed by the users

The steps that I will take in this project:

- **Creating the database**
- **Defying the Entities in IntelliJ IDEA**
- **Create DTOs, Services, and Controllers**
- **Create Repositories**
- **Test Endpoints**

Actions to be taken

- When you open the website, you are prompted to create an account or log in. You put in your first and last name, an email address , phone number, and a password to create an account, while to login you can put your email address and phone number.
- Users can use the website to check , select and purchase the books they want Additionally, they can check some basic information about the books (author, genre and availability within the bookstore).
- While employees after login they can be assigned to handle orders, check inventory and add new books.

Entities and Attributes in the Database

Entity 1 : Book

BookID (unique attribute) – Gives an unique number for each book.

BookTitle – Gives the full title of the book.

BookAuthor – Gives the author that has written the book.

BookGenre – Shows the genre the book belongs to.

BookLanguage – Gives the language the book is in.

Entity 2 : User

UserID (unique attribute) – Gives an unique number for each user.

UserFullName (composite attribute) – Shows the full name of the user.

 UserFName – Gives the first name.

 UserLName – Gives the last name.

UserAge – Gives the age.

UserPhoneNumber – User's phone number.

Entity 3: Employee

EmpID (unique attribute)

EmpName (composite attribute) - Shows the full name of the user.

 EmpFName - Gives the first name.

 EmpLName - Gives the last name.

EmpRole – It shows the role of the employee in the library.

EmpPhoneNum – Employee's Phone number.

Entity 4: Orders

OrderID – (unique attribute)

OrderDate – It shows when the order of the book was placed

Entity 5: Orders_items

Quantity – How much items(books) were ordered.

Entity 6: Ratings

RatingID – (unique attribute)

Score – It shows when the score of the book that was placed by the users

Comment – it shows the comments that the users leave under a book

RatedAt – it shows what time the book was rated at

Relationships between entities

User --- Orders:

One user can place many orders.
Each order belongs to exactly one user

Employee ↔ Order

An employee can handle 0 to many orders.
Orders have to be handles by at least one employee.

User ↔ Rating:

One user can give 0 to many ratings.
Each rating record belongs to exactly one user.

Book ↔ Rating

A book can receive many to 0 ratings.
Each rating record refers to exactly one book.

Foreign keys:

- Orders entity get UserID as foreign key
- Rating entity gets UserID and BookID as foreign keys